4.3 Trade-off Constraints

4.3.1 Physical Constraint – Camera Height and Pillar Detection Performance

Camera Position	Advantage	Limitation
Higher Camera Position	Provides a better viewing position for object detection	Makes the robot's overall structure appear less balanced

Engineering Decision:

- We prioritized a higher camera position to improve the Pixy Camera's viewing range.
- However, considering the robot's overall structure, we chose a smoother and more controlled driving speed instead of prioritizing maximum speed.
- This helps maintain consistent movement throughout the mission.

Camera Height	Trial 1	Trial 2	Trial 3	Detection Limitation	Overall Performance
235 mm	Failed	Failed	Failed	Detected pillars too late	0/3
255 mm	Partial	Partial	Partial	Failed to detect pillar at 40 cm	0/3
275 mm	Partial	Partial	Partial	Failed to detect pillar at 55 cm	0/3
295 mm	**Successful**	**Successful**	**Successful**	**No detection limitation observed**	**3/3**

- The results showed that camera mounting height influenced the Pixy Camera's ability to detect pillars along the testing route.
- At 235 mm, the camera failed to detect the pillars in all three trials. At 255 mm and 275 mm, the robot could detect nearby pillars but failed to recognise certain distant pillars.
- In contrast, the 295 mm configuration successfully detected and avoided all pillars in all three trials. Therefore, 295 mm was selected as the final camera mounting height.

4.3.2 Trade-off Platform Hardware Constraint – Sensor Coverage vs Hardware Complexity

Sensor Configuration	Advantage	Limitation
One Ultrasonic Sensor	Enough hardware connection	Limited distance information from one side
Two Ultrasonic Sensors	Monitor both left and right sides	Requires an additional MUX and more wiring

Engineering Decision:

- We selected two Ultrasonic Sensors to obtain distance information from both sides of the robot.

- Although this increases hardware and wiring complexity, it provides better environmental awareness for navigation and obstacle detection.

Dual Ultrasonic Sensor Navigation Performance Test

Testing Parameter	One Ultrasonic Sensor	Two Ultrasonic Sensors
Number of Trials	5	5
Getting Stuck	4	0
Successful Navigation	1	5
Success Rate	**20%**	**100%**

Engineering Decisions:

- Based on our testing, we chose **two Ultrasonic Sensors instead of one** because the single-sensor configuration resulted in the robot **getting stuck in 4 out of 5 trials**, **achieving only a 20% navigation success rate**.
- In comparison, the **dual-sensor configuration completed all five trials without getting stuck**, **achieving a 100% success rate**. This demonstrated that monitoring both the left and right sides improved the robot's ability to detect obstacles and navigate around them.
- Although using two sensors increases hardware complexity and requires additional wiring, we prioritized navigation reliability over a simpler hardware configuration.
- To overcome the EV3's limited sensor input ports, we purchased an additional EV3 Multiplexer (MUX) through Amazon, allowing both Ultrasonic Sensors to connect through a single sensor input port.

4.3.3 Trade-off Component Constraint – Component Size vs Educational Accessibility

Consideration	Advantage	Limitation
LEGO Mindstorms EV3 Platform	Familiar, accessible and suitable for educational learning	Relatively large components occupy more chassis space
Open Hardware Platform	More flexibility in chassis space	Would require a different learning and development platform

Engineering Decision:

- Although the EV3 Brick, motors and sensors occupy considerable space, we decided to continue using the LEGO Mindstorms EV3 platform.
- This decision supports our educational concept of making robotics more accessible and encouraging more learners to explore robotics through a familiar educational platform.

4.3.4 Trade-off Computational Constraint – Programming Performance vs Development Complexity

Programming Environment	Advantage	Limitation
LEGO Mindstorms EV3	Easier to learn and develop	Slower program execution in our testing
Clev3R	Better program execution performance in our testing	More challenging to learn and develop

Engineering Decision:

- We initially tested the LEGO Mindstorms EV3 programming environment but found its execution performance insufficient for our requirements.
- Therefore, we switched to Clev3R despite its higher development complexity.
- We accepted the additional learning effort to achieve better program execution performance for our robot.

Comparative Analysis Testing of Sensor Data Acquisition Rate

Technical Reference:

According to the OFDL EV3 CLEV3R Modules developed by OFDL Taiwan, converting EV3-G functions into BASIC can achieve approximately 3 times faster program execution and sensor data reading. This provides a technical reference for the performance advantages of CLEV3R-based programming.

Parameter	Mindstorms EV3	Clev3R
Test Program	ReadSensorValue	ReadSensorValue
Test Duration	1 second	1 second
Total Data Readings	300	968
Data Acquisition Rate	300 readings/s	968 reading/s

- Our experimental results showed that Clev3R achieved approximately 3.23 times more sensor readings per second than LEGO Mindstorms EV3.
- This result is consistent with the performance improvement described by OFDL Taiwan.
- Although Clev3R requires additional learning and development effort, we selected it to achieve a higher sensor data acquisition rate for our robot.